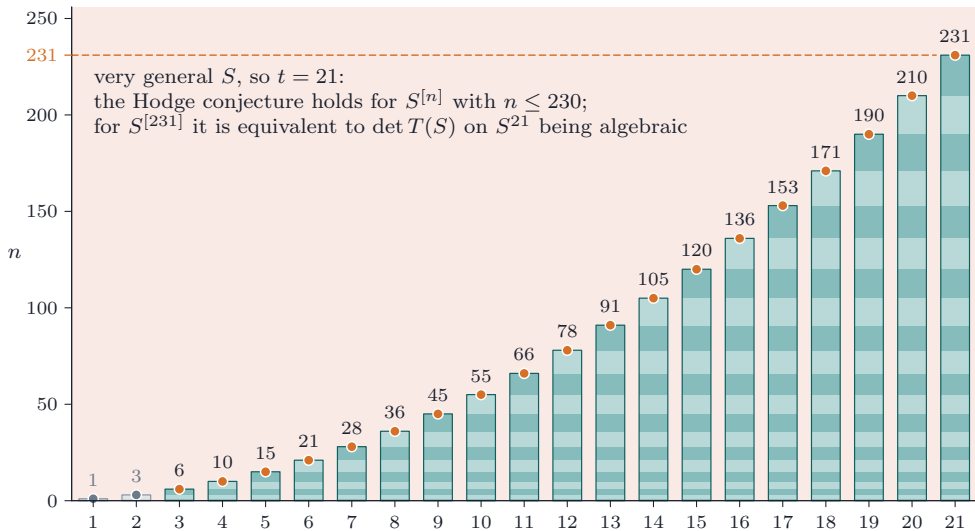




$t$ : the number of distinct part sizes of a partition of  $n$ ; for  $S$  with  $E = \mathbb{Q}$ ,  $t = \dim T(S)$

- $n < t(t+1)/2$ : the Hodge conjecture holds for  $S^{[n]}$ ; bar  $t$  is cut into the parts  $1, 2, \dots, t$
- $n = t(t+1)/2$ : the Hodge conjecture for  $S^{[n]} \Leftrightarrow \det T(S)$  on  $S^t$  is algebraic  $\Leftrightarrow$  it holds for every  $S^k$
- $n > t(t+1)/2$ : open; it follows from the algebraicity of  $\det T(S)$  on  $S^t$
- $t \leq 2$  does not occur for a K3 surface with  $E = \mathbb{Q}$